3.8

ADULTS

Respiratory Tract Infections

Algorithm for Diagnosing Community-Acquired Pneumonia (CAP)

1. Is there pneumonia?

Patient needs to fulfil at least **TWO of A** and **ONE of B**.

A. Clinical criteria

1. Acute onset of fever (T≥38°C)
2. Recent onset of cough or worsening of cough (< 2 weeks duration)
3. New onset of increased sputum production or a change in the character of the sputum (yellowish or greenish)
4. Recent onset of progressively increased dyspnoea
5. Acute onset of pleuritic chest pain
6. Auscultatory findings suggestive of pneumonia on physical examination which include bronchial breath sounds, increased vocal resonance, crepitations
7. Raised total white cell counts (>11 x 10^9/L)
8. For those with pleural effusion, pleurocentesis confirms the presence of neutrophilic inflammatory exudate.

B. Imaging criteria

1. Radiographic image (CXR or CT thorax) shows evidence of lung field infiltrates
2. For patients who have a normal CXR at presentation but have signs and symptoms suggestive of pneumonia, a diagnosis of pneumonia could still be confirmed in retrospect if a repeat chest imaging demonstrates evidence of pneumonia within 48 hours after admission

2. Essential investigations

a. Chest x-ray **b.** Sputum for culture **c.** Blood culture (in moderate to severe CAP)

3. Assess severity according to CURB-65 score

The CURB-65 score is based upon five easily measurable factors from which its name is derived. 1 point is assigned for each criterion that is met.

CURB-65 Score

1. **C**onfusion (based upon a specific mental test or new disorientation to person, place, or time)
2. **U**rea > 7 mmol/L
3. **R**espiratory rate ≥ 30 rpm
4. **B**lood pressure (SBP < 90 mmHg; DBP ≤ 60 mmHg)
5. Age > **65** years old

Scoring	
0-1	Home antibiotics unless the patient has other comorbidity or difficult social circumstances.
2	Admit to hospital.
3	Admit to hospital and consider ICU referral.

Community-Acquired Pneumonia (CAP)

Common organisms	Total duration of antimicrobials	Risk factors for melioidosis
S. pneumoniae, Klebsiella pneumonia, H. influenzae, Moraxella catarrhalis, Legionella sp., Mycoplasma pneumoniae, Chlamydophila (pneumoniae and psittaci), respiratory viruses. *S aureus* may also cause CAP but is uncommon.	**Beta-lactams:** 5-7 days **Azithromycin:** 3 days **<u>If Legionella is suspected:</u>** **Azithromycin:** 5 days **Doxycycline:** 7-10 days (up to 21 days)	Diabetes mellitus with poor glycaemic control Renal dysfunction Occupational or recreational exposure

CHAPTER 3.8	RESPIRATORY TRACT INFECTIONS	
Infections and Likely	**Suggested Treatment**	**Notes**

Organisms	Preferred	Alternative	
Community-Acquired Pneumonia (CAP)			
Low severity CURB-65 = 0-1 or CRB 65 = 0 If hospitalised due to other reasons	Amoxicillin 500-1000 mg PO q8H **OR** Co-amoxiclav 625 mg PO q8H ± Doxycycline 200 mg PO single dose, then 100 mg PO q12H for 5-7 days **OR** Azithromycin 500 mg PO q24H for 3 days Same as above but consider IV route	Cefuroxime 500 mg PO q12H ± Doxycycline 200 mg PO single dose, then 100 mg PO q12H for 5-7 days **OR** Azithromycin 500 mg PO q24H for 3 days	• **Moderate risk of penicillin allergy (delayed rash which is not urticarial or DRESS / SJS / TEN):** Replace Co-amoxiclav with Cefuroxime 500 mg PO q12H **OR** if unable to tolerate oral therapy: Ceftriaxone ± Metronidazole • **High risk Penicillin/Cephalosporin allergy: History suggestive of high risk (e.g., anaphylaxis, urticaria, angioedema, bronchospasm, DRESS / SJS / TEN):** Consult ID/Respiratory Medicine Team
Inpatient (CURB > 2) **Moderate severity** CURB 65 = 2, (9% mortality or CRB 65 = 1 - 2	Co-amoxiclav 1.2 g IV q8H **OR** Cefuroxime 1.5 g IV q8H ± Azithromycin 500 mg	Ceftriaxone 2 g IV q24H ± Azithromycin 500 mg PO/IV q24H for 3 days **OR** Doxycycline 100 mg PO	• Hospitalisation is recommended. • **Moderate risk of penicillin allergy (delayed rash which is not urticarial or DRESS / SJS / TEN):** Same as above. • **High risk Penicillin/Cephalosporin allergy: History suggestive of high risk (e.g., anaphylaxis, urticaria, angioedema, bronchospasm, DRESS / SJS / TEN):** Consult

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Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
	PO/IV q24H for 3 days **OR** Doxycycline 100 mg PO q12H for 5-7 days	q12H for 5-7 days	ID/Respiratory Medicine Team • **If MRSA is suspected:** Add Vancomycin 15 mg/kg (max 1 g) IV q8-12H. • **If melioidosis is suspected:** Use Meropenem or Ceftazidime.
High severity CURB 65 = 3-5, (15 – 40% mortality) or CRB = 3-4	Ceftriaxone 2 g IV q24H + Azithromycin 500 mg IV q24H	Cefepime 2 g IV q8H + Azithromycin 500 mg IV q8H	• **Note:** Consider Meropenem if melioidosis is suspected. • **Moderate risk of penicillin allergy (delayed rash which is not urticarial or DRESS / SJS / TEN):** Refer ID/Respiratory Medicine Team • **High risk Penicillin/Cephalosporin allergy: History suggestive of high risk (e.g., anaphylaxis, urticaria, angioedema, bronchospasm, DRESS / SJS / TEN):** Refer ID/Respiratory Medicine Team • Patients should be managed in HDU or ICU. • For those with septicaemia, consider addition of aminoglycoside. • **If MRSA is suspected:** Add Vancomycin 15 mg/kg (max 1 g) IV q8 - 12H • **Risk factors for melioidosis:** Uncontrolled diabetes and exposure history

Hospital-Acquired Pneumonia (HAP) and Ventilator-Associated Pneumonia (VAP)

Definition

- **HAP:** Pneumonia occurring ≥ 48 hours after hospital admission that was NOT incubating at the time of the hospital admission.
- **VAP:** Pneumonia occurring ≥ 48 hours after endotracheal intubation.

Causative organisms

- *Pseudomonas aeruginosa, Acinetobacter baumannii,* various gram-negative organisms and MRSA.
- Atypical organisms, e.g., Mycoplasma sp, Chlamydophila sp or Legionella are rare.

Risk factors for multidrug-resistant pathogens

- Prior intravenous antibiotic use within 90 days
- Septic shock at time of VAP
- Acute respiratory distress syndrome (ARDS) preceding VAP
- Five or more days of hospitalisation prior to the occurrence of VAP

- Acute renal replacement therapy prior to VAP onset

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Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
Hospital-Acquired Pneumonia (HAP)/Ventilator-Associated Pneumonia (VAP)			
Early onset HAP/VAP (2-4 days of admission/ intubation)	Co-amoxiclav 1.2 g IV q8H for 5-7 days + Azithromycin 500 mg PO q24H for 3 days	Ceftriaxone 2 g IV q24H for 5-7 days + Azithromycin 500 mg PO q24H for 3 days ± Metronidazole 400 mg PO q8-12H **OR** Metronidazole 500 mg IV q8H*	• **Moderate risk of penicillin allergy (delayed rash which is not urticarial or DRESS / SJS / TEN):** Replace Co-amoxiclav with Cefuroxime 500 mg PO q12H **OR** if unable to tolerate oral therapy: Ceftriaxone ± Metronidazole • **High risk Penicillin/Cephalosporin allergy: History suggestive of high risk (e.g., anaphylaxis, urticaria, angioedema, bronchospasm, DRESS / SJS / TEN):** Consult ID/Respiratory Medicine Team • *If suspected aspiration or recent thoraco-abdominal surgery
Late onset HAP/VAP (5 days or more of admission/ intubation) Non-severe pneumonia and did not receive IV antibiotics in the	Co-amoxiclav 1.2 g IV q8H for 7 days	Ceftriaxone 2 g IV q24H for 7 days	• **High risk Penicillin/Cephalosporin allergy: History suggestive of high risk (e.g., anaphylaxis, urticaria, angioedema, bronchospasm, DRESS / SJS / TEN):** Refer ID/Respiratory Medicine Team • Duration of antibiotics could be shortened to 7 days in patients without immunodeficiency, cystic fibrosis, empyema,

Infections and Likely Organisms	Suggested Treatment		Notes
	Preferred	**Alternative**	
preceding 90 days Severe pneumonia **OR** received IV antibiotics in the preceding 90 days	Cefepime 2 g IV q8H for 7 days	Piperacillin/Tazobactam 4.5 g IV q6H for 7 days **OR** Meropenem 1 g IV q8H for 7 days	lung abscess, cavitation or necrotising pneumonia and with a good clinical response to therapy • Longer duration may be indicated depending on clinical, radiological and laboratory parameters. • To de-escalate antibiotics according to culture and sensitivity results. • **If patient is colonised with MRSA:** add Vancomycin 20 mg/kg (max 2 g) single dose, then 15 mg/kg (max 1 g) IV q8-12H. • **If patient is colonised with Carbapenem-resistant *Acinetobacter baumannii* (CRAB):** add Polymyxin B 2.5 mg/kg IV single dose, then 1.5 mg/kg IV q12H over 1 hour (to start 12 hours after loading dose).
Aspiration Pneumonia			
• Most aspiration is sterile and requires physiotherapy, not antibiotics. • If infection develops, anaerobes and mouth Streptococci (e.g., *milleri/anginosus*) should be treated. • Antibiotics are **NOT** indicated for aspiration (chemical) pneumonitis.			
Community-acquired	Co-amoxiclav 625 mg PO q8H	Cefuroxime 750 mg IV q8H	• Minor degree of aspiration does not require antibiotics. • With acute intestinal obstruction:

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Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
	OR Co-amoxiclav 1.2 g IV q8H	**OR** Metronidazole 500 mg IV q8H	Add Metronidazole 500 mg IV q8H.
Hospital-acquired	Piperacillin/Tazobactam 4.5 g IV q6H		
Lung Abscess			
• Those with cavities < 2 cm are also known as necrotising pneumonia or lung gangrene. • It is often secondary to aspiration - hence, anaerobes are important. • Other causes include Gram-positive cocci, *Klebsiella pneumoniae, S. milleri / anginosus* and rarely, Nocardia sp. • May change to oral regime once clinical improvement is seen.			
Empirical, Community-acquired	Co-amoxiclav 1.2 g IV q8H **Followed by oral step-down:** Co-amoxiclav 1 g PO q8H	Ceftriaxone 2 g IV q24H + Metronidazole* 500 mg IV q8H **OR** Clindamycin 600 mg IV/PO q6H	• **Duration of treatment:** • Drained abscess: may require 2 – 4 weeks of antibiotics. • Undrained abscess/empyema: may require 4 – 6 weeks of antibiotics. • Lung empyema: Attempts should be made to drain the collection.

CHAPTER 3.8	RESPIRATORY TRACT INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
			• *Add Metronidazole in cases of lung abscess when aspiration is suspected. • Use Meropenem 1 g IV q8H if melioidosis is suspected. • Risk factors for melioidosis: • Diabetes mellitus with poor glycaemic control • Renal dysfunction • Occupational or recreational exposure • Add Vancomycin 15 mg/kg (max 1 g) IV q8-12H for MRSA coverage if penetrating injury. • If *Staphylococcus aureus* is isolated, streamline antibiotics to: • Cloxacillin 2 g IV q4-6H **OR** Cefazolin 2 g IV q8H, for 4 – 6 weeks • May change to oral therapy to complete the duration once patient has improved. • Cefalexin 1 g PO q6H **OR** Cloxacillin 2 g PO q6H

Bronchiectasis

Immunisation
Offer annual influenza immunisation and polysaccharide pneumococcal to all patients with bronchiectasis.

Common organisms causing acute exacerbation of bronchiectasis
Pseudomonas aeruginosa, Haemophilus influenzae, Streptococcus pneumoniae, Moraxella spp., Catarrhalis spp.*, S. aureus.*

Bronchiectasis Severity Index		
Mild/Moderate	**Severe**	**Very Severe**
Exacerbation can be controlled with oral antibiotic treatment	Exacerbation requires intravenous antibiotic treatment or hospitalization. Those that present with at least one of the following conditions are also considered severe. • Exacerbated acute or chronic respiratory failure, • Significant deterioration in oxygen saturation, • High temperature or other criteria for sepsis, • Frank haemoptysis or significant	Exacerbations that present with hemodynamic instability, altered level of consciousness, or need for admission to an intensive or intermediate care unit

	deterioration in lung function.	

CHAPTER 3.8	RESPIRATORY TRACT INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
Bronchiectasis			
Mild – moderate	Co-amoxiclav 625 mg PO q8H	Cefuroxime 500 mg PO q12H	• **Moderate risk of penicillin allergy (delayed rash which is not urticarial or DRESS / SJS / TEN):** Replace Co-amoxiclav with Cefuroxime 500 mg PO q12H • **High risk penicillin / cephalosporin allergy (History suggestive of high risk (e.g., anaphylaxis, urticaria, angioedema, bronchospasm, DRESS / SJS / TEN):** Refer ID/Respiratory Medicine Team • Good practice: trace previous sputum bacteriology to guide the antibiotic choice
Severe or failed oral therapy If patient is in sepsis	Ceftazidime 2 g IV q8H **Add Aminoglycoside to above:** Gentamicin 5 mg/kg IV	Piperacillin/Tazobactam 4.5 g IV q6H **OR** Cefepime 2 g IV q8H **Add Aminoglycoside to above:** Gentamicin 5 mg/kg IV	• **Moderate risk of penicillin allergy (delayed rash which is not urticarial or DRESS / SJS / TEN):** Replace Piperacillin/Tazobactam with Cefepime • **High risk penicillin / cephalosporin allergy (History suggestive of high risk (e.g., anaphylaxis, urticaria, angioedema, bronchospasm, DRESS / SJS / TEN):** Refer ID/Respiratory Medicine Team

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Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
	q24H **OR** Amikacin 15 mg/kg IV q24H	q24H **OR** Amikacin 15 mg/kg IV q24H	
Bronchiectasis (Chronic Suppressive Treatment)			
Antibiotic treatment may only be justified for patients with frequent exacerbations (≥ 3 per last 12 months; or ≥ 2 per last 12 months with significant comorbidities).			
Regular maintenance treatment	Azithromycin 250-500 mg PO 3 times weekly	Erythromycin Ethyl Succinate 400 mg PO q12H	
Bronchiectasis (Eradication Treatment)			
Pseudomonas aeruginosa	**Initial 2 weeks treatment:** Ciprofloxacin 500-750 mg PO q12H	**Initial 2 weeks treatment:** Ceftazidime 2 g IV q8H* ± Amikacin 15 mg/kg IV q24H **OR** Gentamicin 5 mg/kg IV	• *To treat based on sensitivities. • If the strain is resistant to amikacin and gentamicin, to consider Nebulised Colistin 2 MU q12H.

CHAPTER 3.8	RESPIRATORY TRACT INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
		q24H	
	Followed by a 3-month course of nebulised: Amikacin 250-500 mg q12H **OR** Gentamicin 160 mg q12H	**Followed by a 3-month course of nebulised:** Amikacin 250-500 mg q12H **OR** Gentamicin 160 mg q12H	

Acute Exacerbation of Chronic Obstructive Pulmonary Disease (AECOPD)

A. Anthonisen Classification on Severity of AECOPD	B. Global Initiative for Chronic Obstructive Lung Disease (GOLD)
Type 1: Fulfilled ALL three criteria below (antibiotics are indicated)	• Patient is intubated
• Increased purulence in sputum	

- Increased dyspnoea
- Increased sputum volume

Type 2: Fulfilled TWO of the above criteria (consider antibiotics)

CHAPTER 3.8	RESPIRATORY TRACT INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
Acute Exacerbation of Chronic Obstructive Pulmonary Disease (COPD)			
Outpatient	Co-amoxiclav 625 mg PO q8H for 5 – 7 days ± Doxycycline 200 mg PO single dose, then 100 mg PO q12H for 5 – 7 days **OR** Azithromycin 500 mg q24H for 3 days	Cefuroxime 500 mg PO q12H for 5 -7 days ± Doxycycline 200 mg PO single dose, then 100 mg PO q12H for 5 – 7 days **OR** Azithromycin 500 mg q24H for 3 days	• Not all COPD exacerbations require antibiotic therapy. • Antibiotic may be considered if the patient has one or more of the following symptoms or scenarios: • Purulent sputum production • Fever • Exacerbation which requires hospitalisation
Inpatient	Co-amoxiclav 1.2 g IV q8H for 5-7 days **OR** Cefuroxime 1.5 g IV q8H	Ceftriaxone 2 g IV q24H for 5-7 days ± Azithromycin 500 mg IV	

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Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
	for 5-7 days ± Azithromycin 500mg IV q24H for 3 days **OR** Doxycycline 100mg PO q12H for 5-7 days	q24H for 3 days **OR** Doxycycline 100 mg PO q12H for 5-7 days	
Viral Infections			
Varicella zoster pneumonia	Aciclovir 10 mg/kg IV q8H for 7 days **OR** Valaciclovir* 1000 mg PO q8H for 5 days		• Pregnancy (3rd trimester): • Aciclovir 800 mg PO 5 times daily **OR** 10 mg/kg IV q8H for 5 days. • *Valaciclovir is not available in UMMC Formulary.
Influenza Post-exposure prophylaxis for 'at risk' population	Oseltamivir 75 mg PO q24H for 7 days		
Influenza	Oseltamivir 75 mg PO q12H for 5 days		• Treat at risk population if suspected influenza:

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Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
To treat 'at risk' population Initiate treatment within 48 hours of influenza symptom onset			• Chronic respiratory disease (including COPD and asthma) • Significant cardiovascular disease (not hypertension) • Severe immunosuppression • Chronic neurological, renal, or liver disease • Diabetes mellitus • Morbid obesity (BMI>40) • Persons of any age who are hospitalised with influenza, regardless of illness duration prior to hospitalisation • Children younger than 6 months and adults ≥65 years • Pregnant women and those within 2 weeks postpartum • Antibiotics should not routinely be given to patients with uncomplicated influenza. • Antibiotics should not be given to prevent bacterial complications. • Investigate and empirically treat bacterial coinfection in patients who deteriorate after initial improvement, particularly in those treated with antivirals.

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Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
			• Proper sample collection: IF/viral panel PCR.
Asthma Exacerbation			
Clinical practice guidelines recommend AGAINST empiric antibiotic therapy for the treatment of an asthma exacerbation because most respiratory infections that trigger an exacerbation of asthma are viral rather than bacterial			

Switching Intravenous to Oral Therapy

Criteria
• May start as early as on day 3 of antimicrobial.
• Patient is afebrile, temperature < 37.5°C.
• Clinically improving and/or inflammatory markers showing downward trend.
• No contraindications to take orally.

Oral antibiotic options for conversion from intravenous (IV) route

Note: Choose based on the result of culture and sensitivity (C&S) if available.

- Co-amoxiclav 625 mg q8H* ± Azithromycin / Doxycycline
  (*Use Co-amoxiclav 1 g q8H for lung abscess/empyema)
- Cefuroxime 500 mg q12H ± Azithromycin / Doxycycline
- Moxifloxacin 400 mg q24H **(only if the above are contraindicated)**

Duration of treatment

- 5 to 7 days for most infections.
- May extend treatment to 10 days in *Pseudomonas aeruginosa* and *Legionella* infections.